Inequalities — problem 7.87

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Let a, b, c be the side lengths of a triangle, and let s_a, s_b, s_c be the lengths of the medians. D is the diameter of the circumcircle. Prove that

$$\frac{a^2+b^2}{s_c}+\frac{b^2+c^2}{s_a}+\frac{c^2+a^2}{s_b}\leq 6D.$$